

PROJECT REPORT: AI IT HELPDESK AGENT

2. RAG Semantic Retrieval: Searches Knowledge Base SOPs using TF-IDF & Cosine Similarity.

Track #4: Diagnoses Technical Issues & Recommends Steps Using Knowledge Base

1. Project Overview

1.1 Project Title

AI IT Helpdesk Agent: Autonomous Diagnostic Reasoning, RAG Knowledge Retrieval & Tools

1.2 Problem Statement

Enterprise IT helpdesks face overwhelming volumes of repetitive Level-1 support requests

- Automated Tools: search_knowledge_base, run_system_diagnostic, create_it_ticket,

1.3 Brief Description of the Project

The AI IT Helpdesk Agent is an end-to-end agentic solution combining a Vector RAG Engine,

2. Objectives & Proposed Solution

2.1 Project Objectives

- Automate technical diagnosis for VPN, BSOD, Wi-Fi, Outlook, and account lockouts.

2.2 How the Agentic AI Solution Works

1. User Query Ingestion: Receives technical complaint via Web Chat or CLI.

2.3 Key Features

- Semantic RAG Vector Index pre-loaded with IT troubleshooting SOPs.

3. Implementation & Results

3.1 Technologies / Tools Used

- Core Runtime: Python 3.14

- Hallucination Risk: Controlled via strict RAG thresholds & tool observation checks.

3.2 Working Process & Code Architecture

ai_it_helpdesk_agent/

RAG knowledge retrieval, and tool execution, reducing Level-1 IT support resolution times

3.3 Output Screenshots & Interface Verification

- ServiceNow / Jira Service Desk REST API integration.

- Web App URL: <http://localhost:8000> (Active & Running Live)
Console Encoding: Sanitized Windows CP 1252 prints while keeping rich Web UI.

3.4 Results Achieved

- 100% Automated Test Pass Rate: 5/5 unit tests passed in 0.008 seconds.

4. Conclusion & Future Scope

OS Native Probes using psutil and subprocess ping commands.

4.1 Project Conclusion

The AI IT Helpdesk Agent successfully demonstrates automated diagnostic reasoning.

3. Python Software Foundation. "Standard Library Documentation." <https://docs.python.org/3/>

4.2 Challenges Faced

- Environment Memory Constraints: Resolved via zero-dependency native REST server.

2. Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks."

4.3 Future Enhancements

- Active Directory / LDAP Binding for live account unlocking.

4.4 References

1. Yao, S., et al. (2022). "ReAct: Synergizing Reasoning and Acting in Language Models."